Zeros and Ones | Sahara

Health-a-thon 2026

About the Team / Organisation

A. **Team Zeros and Ones (WeDevIT).** Shaik Abdul Basith (technical lead: website and the AI that writes down the conversation) and Shaik Muhammad Awaiz (Android app and server) built Sahara. Shirin Ayub (medical student) designed how patients log symptoms and researched caregiver stress. Legal advice was sought: inputs and refining the goals-of-care wording and format.

B. **Doctor partners.** Dr Sharada Vinod Kutty and Dr Sujay Halkur Shankar are pulmonologists (lung specialists) who treat many lung-cancer patients; Dr Sujay also works in palliative care (care that eases pain and distress in a serious illness). They chose this problem from their own clinics, wrote the clinical specification (the medical rules the app follows), set the order of the conversation checklist, and tested the app with us, most recently in the 3 Nov team meeting.

Problem Statement (use case specific)

Goals-of-care wishes are lost when they matter most. In a goals-of-care discussion (advance care planning), the doctor asks a seriously ill patient and family what treatment they want or refuse near the end of life: CPR (restarting the heart), a ventilator, ICU, dialysis, where to die. It is rarely written down as a clinical document that can be found later. Example: a 62-year-old woman with advanced breast cancer tells her doctor her wishes; two months later she is unconscious in the emergency department, her children disagree about CPR, and nobody can find what she said.

Evidence: about 15.6 lakh new cancer cases in India in 2024 (GLOBOCAN); 98% of patients with metastatic (spread) cancer had unmet palliative-care needs (Patil et al., 2024); 88% of surveyed urban patients wanted to choose their own end-of-life treatment (IndiaSpend, 2019); only 11.5% of doctors in one hospital were trained in advance care planning (Jacob et al., 2026). Families carry the load too: about 38% of family caregivers report high emotional stress and about 1 in 5 has clinical burnout (Cleveland Clinic and other surveys).

Users and setting: patients receiving palliative care and their families, at home; their palliative care physician, oncologist or pulmonologist, in the clinic (as an extension plugin to their electronic medical records system); and the emergency physician, who has only minutes to decide and needs the patient's wishes immediately.

AI Solution Description

Sahara is a palliative-care app (Android and website). In the clinic the doctor leads the goals-of-care discussion; Sahara listens and writes. After the patient ticks "I agree", the doctor taps Record. Google's Gemini AI transcribes the talk (turns speech into text), labels who said what, and places each answer, word for word, on the goals-of-care document, a clinical document. A checklist in the doctors' order ticks itself and lists any topic not yet discussed. The doctor corrects it; doctor and patient or family sign on the phone, and the document locks: any later edit cancels the signatures. It opens from the patient's emergency QR code (a barcode any phone camera can scan). The AI only writes things down: the doctor makes every clinical decision.

Technology and Tools Used

Android (Kotlin, Jetpack Compose); website (Next.js, Vercel). **Supabase** is the server: sign-in, a Postgres database with row-level security (each family sees only its own records), and server functions that keep the AI keys off the phone. **Google Gemini** (backup models when busy) transcribes, fills the document and answers "Ask your documents" with the line it used. With no internet, **OpenAI Whisper** transcribes on the phone itself; if that fails, the doctor types. Google ML Kit: 5 languages and read-aloud on the phone.

Key Features of the solution

- **Goals-of-care discussion (advance care planning):** consent, record, transcribe, checklist in the doctors' order, fill the clinical document, sign and lock.
- **Emergency QR code and SOS:** goals of care, contacts and medicines in one scan for the emergency physician.
- **Symptom diary:** 0–10 scores covering the ESAS-r symptoms (pain, tiredness, breathlessness, nausea, appetite, mood) and a pain body map; charts for the doctor, most severe first.
- **Distress check:** NCCN Distress Thermometer and Problem List (NCCN permission being sought).
- **Caregiver wellbeing check (new):** family members rate their own stress, tracked like a symptom; if it worsens, Sahara points to counselling, Tele-MANAS (14416), respite care and caregiver support groups. It supports; it does not diagnose.
- **Medicines and reports:** one tick per dose, reminders, prescription photo, tubes-and-drains tracker; ask a question about a hospital report.
- **Help nearby and family:** palliative centres, ambulances, support groups; invited family share the record. Five languages, read aloud.

Build Sprint Progress

- **Weeks 1–2 (9–22 Oct):** shortlisted 9 Oct; native Android app, on Google Play closed testing from 14 Oct; map of nearby palliative centres; medicine organiser with prescription photo; symptom tracker.
- **Week 3 (23–29 Oct):** pain body map; hospital survey live 27 Oct; 28 Oct team meeting (9 decisions); "Ask your documents"; Dr Sharada's clinical specification on 29 Oct.
- **Week 4 (30 Oct–5 Nov):** advance care planning (goals of care) and emergency QR code; Telugu, Urdu, Kannada; legal advice 31 Oct; Dr Sujay tested the whole app 3 Nov; goals-of-care recording with Gemini onto the clinical document; signed, locked PDF.
- **How it changed:** in Round 1 the AI drafted a "Care Note"; our doctors asked that the doctor lead and the AI only write it down, on the clinical document. 7 Nov: caregiver wellbeing check added.

User Validation and AI Evaluation

- **Automatic tests:** 877 Android and 594/594 website tests pass (they re-check every feature after each change).
- **Goals of care:** 8 made-up conversations (6 languages, a change of mind, a family disagreement): every answer on the right line of the document, 8 of 8.
- **Ask your documents:** 4 of 4 questions on a sample hospital report answered with the right source line.
- **Dr Sujay, 3 Nov meeting:** asked for a guide on what to talk about next → checklist in the doctors' order (consent, patient, family, CPR, dialysis…) with a "not yet discussed" prompt; 4 doses a day → 4 ticks.
- **Legal advice, 31 Oct:** inputs and refining the goals-of-care wording and format.
- **Not tested yet:** no real-patient pilot; Gemini not yet measured on real Hindi or Telugu recordings; on-phone speech weak for Telugu, Kannada (typing offered).

Safety and Responsible AI

- **Doctor in charge:** the doctor leads, corrects and signs every goals-of-care document; the AI never diagnoses or suggests treatment, and lists missing topics instead of guessing.
- **Consent and privacy:** recording starts only after the patient ticks "I agree"; Gemini is used only with that consent. Family members join by invite and see only their patient.
- **Tamper-evident documents:** signed documents are sealed with a SHA-256 fingerprint, so any change cancels the signatures; earlier versions stay visible.
- **Inclusion:** five languages, read aloud, large text, thumb signature; typing when speech fails. Demos use made-up patients only.

Deployment and Sustainability Plan

- **Pilot:** 60–90 days in one palliative-care clinic with our doctors; patients consent as India's data-protection law (DPDP Act 2023) requires. We count talks signed and how fast wishes are found.
- **Integration:** works on its own, no hospital software needed; can sit beside the clinic's electronic medical records as a plug-in; a FHIR export (the standard format hospital systems read) is ready.
- **Ambient listening (next):** today Sahara records on the doctor's phone; next, room or lapel microphones in clinics capture the discussion hands-free, with the same consent.
- **Cost (team estimate):** servers about ₹3,900 a month (Supabase, Vercel); Gemini about ₹2 per 30-minute talk; about ₹1,600 per clinic a month if 10 clinics share it. Free for patients; clinics or NGOs pay ₹1,500–2,000 a month.

Supporting Links

- MVP: sahara.wedevit.in · Android app on Google Play
- Video Demo Walkthrough: link to be added
- Other Supporting Links: code (private, access on request)